Numerical Analysis homework 2

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I.

(Due to the complicated fractions and equations, it's too troublesome to type them in LaTeX. Therefore, I solve them by hand.)

I ∵ Linear interpolation of f ∴ We have $p_1(f;x) = f(x_0)\frac{x-x_1}{x_0-x_1} + f(x_1)\cdot\frac{x-x_0}{x_1-x_0}$

∴ $f(x) - p_1(f;x) = \frac{1}{x} + f(x_0)\frac{x-x_1}{x_1-x_0} - f(x_1)\frac{x-x_0}{x_1-x_0} = \frac{x_1x_0(x_0-x_1)+x x_0(x-x_0)-x x_1(x-x_1)}{x\cdot x_0 x_1(x_0-x_1)}$

Simplify it and we have

$$LHS = \frac{x^2 - x(x_0+x_1)+x_0x_1}{x_0x_1\cdot x} = \frac{(x-x_0)(x-x_1)}{x_0\cdot x_1\cdot x}$$

∵ $RHS = \frac{1}{2}\cdot f''(\xi(x))(x-x_0)(x-x_1)$

and $\frac{1}{2}f''(x) = \frac{1}{x^3}$ ∴ $\frac{1}{2}f''(\xi(x)) = \frac{1}{\xi^3(x)}$ ∴ Compare both sides and we have $\xi^3(x) = x_0x_1\cdot x$, namely $\xi(x) = \sqrt[3]{x_0\cdot x_1\cdot x}$

I-a. for specific $x_0=1$. $x_1=2$, then $\xi(x)=\sqrt[3]{2x}$.

I-b. $\xi(x) = (2x)^{\frac{1}{3}}$ is monotonically increasing.

∴ $\min\xi(x) = \xi(x_0) = 2^{\frac{1}{3}}$. $\max \xi(x) = \xi(x_1) = 4^{\frac{1}{3}} = 2^{\frac{2}{3}}$.

$f''(\xi(x)) = \frac{1}{x_0\cdot x_1\cdot x}$ is monotonically decreasing in $[x_0, x_1]$

∴ $\max f''(\xi(x)) = f''(\xi(x_0)) = \frac{1}{1\times 2\times 1} = \frac{1}{2}$.

Figure 1: Problem I

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II.

Refer to the Lagrange formula, the basis $l_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}$.

We've been given $(n+1)$ points to construct a polynomial of degree $2n$, also the polynomial needs to be non-negative in the whole real line. Therefore, we can use squares of $l_k(x)$ to construct the polynomial:

$$l_k(x) = \prod_{i=0, i\neq k}^{n} \frac{(x - x_i)^2}{(x_k - x_i)^2},$$

which still satisfies $l_k(x_i) = 0, \quad i \neq k$ and $l_k(x_k) = 1$, while the degree is $2n$.

So the function we want is

$$p(x) = \sum_{k=0}^{n} f(x_k) l_k(x) = \sum_{k=0}^{n} f(x_k) \prod_{i=0, i\neq k}^{n} \frac{(x - x_i)^2}{(x_k - x_i)^2}$$

III.

III-a. ① Initial case: $k=0$. $f[t] = f(t) = e^t$ ~~n=0~~, which matches $\frac{(e-1)^n}{n!} e^t \Big|_{n=0}$

$k=1$. $f[t, t+1] = \frac{f(t+1) - f(t)}{(t+1) - t} = e^{t+1} - e^t = (e-1)e^t$. still correct.

② Assume that $f[t, t+1, \dots, t+k] = \frac{(e-1)^k}{k!} e^t$ is valid, then we have

$f[t, t+1, \dots, t+k+1] = \frac{1}{k+1}\left(f[t+1, \dots, t+k+1] - f[t, \dots, t+k]\right)$

$= \frac{1}{k+1}\left(\frac{(e-1)^k}{k!} e^{t+1} - \frac{(e-1)^k}{k!} e^t\right) = \frac{(e-1)^{k+1}}{(k+1)!} e^t$,

which matches the case of $n = k+1$

③ Therefore, by induction, $f[t, t+1, \dots, t+n] = \frac{(e-1)^n e^t}{n!}$ is valid.

Figure 2: Problem III-a

III-b

For $t = 0$, we have $f[0, 1, \dots, n] = \frac{(e-1)^n}{n!}$. We need to find out the ξ in $[0, n]$ such that $f[0, 1, \dots, n] = \frac{f^{(n)}(\xi)}{n!}$. Since $f(x) = e^x$, we have $f^{(n)}(x) = e^x$. Therefore, we have $e^\xi = (e-1)^n$. Do *log* on both sides, we have

$$\xi = n \ln(e - 1)$$

.

Roughly, we can calculate that $\ln(e-1) \approx 0.541 > 0.5$.

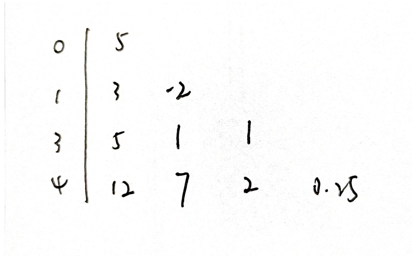
Therefore, $\xi > \frac{1}{2} n$, in other words, located to the **right side of the midpoint**.

IV.

IV-a

Constructing the table below: So we have

$$p_3(f; x) = 5 - 2x + x(x-1) + 0.25x(x-1)(x-2) = 0.25x^3 + 0.25x^2 - 2.5x + 5$$



0	5			
1	3	-2		
3	5	1	1	
4	12	7	2	0.25

Figure 3: table of divided differences

IV-b

We have the approximate derivative of $f(x)$:

$$f'(x) \approx p_3'(f;x) = 0.75x^2 + 0.5x - 2.5 = 0.25(3x^2 + 2x - 10)$$

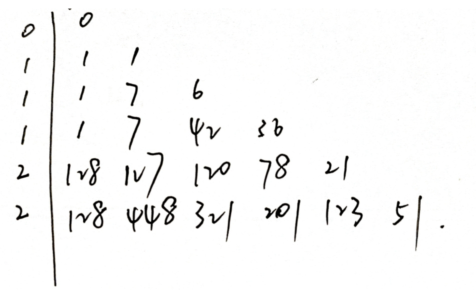
Let $p_3'(f;x) = 0$, we have $x = \dfrac{-1 \pm \sqrt{31}}{3}$. Only $x = \dfrac{-1 + \sqrt{31}}{3} \approx 1.523$ is in the interval $(1, 3)$.

$$\therefore \hat{x}_{min} = 1.523$$

V.

V-a

For $f(x) = x^7$, we have $f'(x) = 7x^6$, $f''(x) = 42x^5$, Constructing the table below:



0	0					
1	1	1				
1	1	7	6			
1	1	7	42	36		
2	128	127	120	78	21	
2	128	448	321	201	123	51

Figure 4: table of divided differences

$$\therefore f[0, 1, 1, 1, 2, 2] = 51$$

V-b

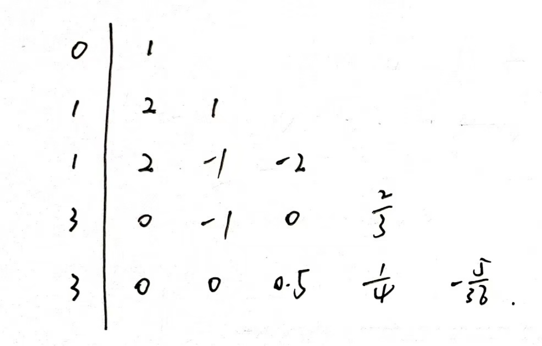
According to **Theorem 2.22**, we have $f[x_0, x_1 \ldots x_n] = \dfrac{1}{n!} f^{(n)}(\xi)$. Here $n = 5$, so we have

$$f[x_0, x_1, x_2, x_3, x_4, x_5] = f[0, 1, 1, 1, 2, 2] = \frac{1}{5!} f^{(5)}(\xi) = 51$$

For $f(x) = x^7$, we have $f^{(5)}(x) = 2520x^2$. So we have

$$\frac{1}{5!} f^{(5)}(\xi) = 21\xi^2 = 51$$

$$\therefore \xi = \sqrt{\frac{17}{7}} \approx 1.558 \in (0, 2)$$



0	1				
1	2	1			
1	2	-1	-2		
3	0	-1	0	$\frac{2}{3}$	
3	0	0	0.5	$\frac{1}{4}$	$-\frac{5}{36}$.

Figure 5: table of divided differences

VI.

VI-a

Based on Hermite interpolation rules, constructing the table below: So we have

$$
\begin{aligned}
f(x) \approx p(f;x) &= 1 + x - 2x(x-1) + \frac{2}{3}x(x-1)^2 - \frac{5}{36}x(x-1)^2(x-3) \\
&= 1 + x(1 + (x-1)(-2 + (x-1)(\frac{2}{3} - \frac{5}{36}(x-3))))
\end{aligned}
$$

$$\therefore f(2) \approx \frac{11}{18} \approx 0.6111$$

VI-b

According to **Theorem 2.7**, we have $R_n(f;x) = \dfrac{f^{(n+1)}(\xi)}{(n+1)!}\Pi_{i=0}^{n}(x - x_i)$.

$$\therefore R_4(f;x) = \frac{f^{(5)}(\xi)}{5!}\Pi_{i=0}^{4}(x - x_i), \quad \xi \in (0,3)$$

$$\because |f^{(}5)(x)| \le M, \quad x \in [0,3]$$

$$\therefore |R_4(f;x)| \le \frac{M}{5!}|x(x-1)^2(x-3)|, \quad x \in [0,3]$$

Now, we have to find out the maximum of $|x(x-1)^2(x-3)|$ in $[0,3]$. The shape of the curve is shown in figure 6.

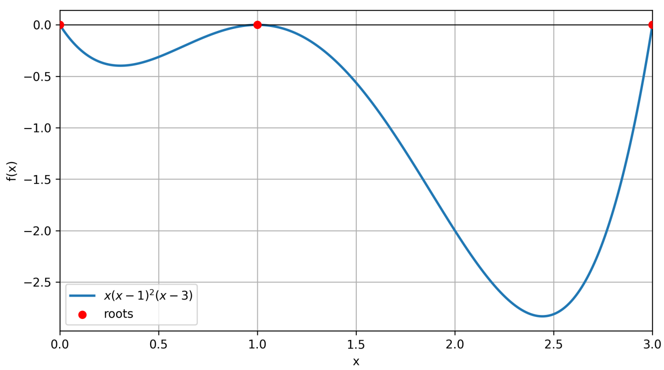


Figure 6: Curve of $y = x(x-1)^2(x-3)$

By observation, we can see that the maximum is roughly located at $x = 2.4$. For $y = x(x-1)^2(x-3) = x^4 - 5x^3 + 7x^2 - 3x$, we have $y' = 4x^3 - 15x^2 + 14x - 3$. Use **Newton's method** in the 1st programming homework to find out the root of $y' = 0$:

```
    NewtonSolver(Function &F) : F(F) {}
    double solve(double x0)
    {
        int M = 5;
        double x = x0;
        int k = 0;
        for (; k < M; k++)
        {
            double u = F(x);
            x -= u / (F.d(x));
        }
        return x;
    }

class Func1 : public Function
{
public:
    double operator()(double x)
    {
        return 4*pow(x,3) -15*pow(x,2) + 14*x - 3;
    }
    double d(double x)
    {
        return 12*pow(x,2) - 30*x + 14;
    }
};

int main()
{
    Func1 func;
    NewtonSolver solver(func);
    double x = solver.solve(2.4);
    cout << x << endl;
    return 0;
}
```

After running the code, we get $x = 2.443$.

$$\therefore |x(x-1)^2(x-3)| \leq 2.8334, \quad x \in [0,3]$$

$$\therefore |R_4(f;x)| \leq \frac{2.8334M}{120} = 0.02361M, \quad x \in [0,3]$$

the maximum possible error

$$\epsilon = 0.02361M$$

VII.

Prove by induction:

① For $k=1$:

$$\Delta^k f(x) = \Delta f(x) = f(x+h) - f(x) = h \cdot \frac{f(x+h)-f(x)}{h} = h \cdot f[x_0, x_1].$$

(for $x_j = x + jh$), which is coherent with the equation.

② Assume that $k=m$ the equation is valid.

namely $\Delta^m f(x) = m!\, h^m f[x_0, x_1, \dots, x_m]$ and we have

$$\Delta^m f(x+h) = m!\, h^m f[x_1, \dots, x_m, x_{m+1}].$$

When $k=m+1$:

$$\begin{aligned}
\Delta^{m+1} f(x) &= \Delta^m f(x+h) - \Delta^m f(x) \\
&= m!\, h^m \left(f[x_1, \dots, x_m, x_{m+1}] - f[x_0, x_1, \dots, x_m]\right) \\
&= m!\, h^m \cdot (m+1)h \cdot \frac{f[x_1, \dots x_m, x_{m+1}] - f[x_0, \dots, x_m]}{(m+1)h} \\
&= (m+1)!\, h^{m+1} \cdot f[x_0, x_1, \dots, x_{m+1}],
\end{aligned}$$

also matches the general equation.

③ Therefore. $\Delta^k f(x) = k!\, h^k f[x_0, x_1, \dots, x_k]$ for all $k = 1, 2, \dots$ is valid.

For the same reason,

$$\begin{aligned}
\nabla^{m+1} f(x) &= \nabla^m f(x) - \nabla^m f(x-h) \\
&= m!\, h^m \left(f[x_0, x_{-1}, \dots, x_{-m}] - f[x_{-1}, x_{-2}, \dots, x_{-m-1}]\right) \\
&= (m+1)!\, h^{m+1} \frac{f[x_0, x_{-1}, \dots, x_{-m}] - f[x_{-1}, x_{-2}, \dots, x_{-m-1}]}{(m+1)h} \\
&= (m+1)!\, h^{m+1} f[x_0, x_{-1}, \dots, x_{-m-1}]
\end{aligned}$$

Similarly, by induction and $\nabla^k f(x) = h^k \cdot k!\, f[x_0, x_{-1}, \dots, x_{-k}]$ is valid.

Figure 7: Problem VII

VIII.

The definition of partial derivative is:

$$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, x_2, \dots, x_i + h, \dots, x_n) - f(x_1, x_2, \dots, x_i, \dots, x_n)}{h}$$

According to it, viewing x_o as a variable,

$$\therefore \frac{\partial}{\partial x_0} f[x_0, x_1, \ldots, x_n] = \lim_{h \to 0} \frac{1}{h}[f(x_0 + h, x_1, \ldots, x_n) - f(x_0, x_1, \ldots, x_n)]$$

Based on the definition of divided difference, we have

$$\frac{\partial}{\partial x_0} f[x_0, x_1, \ldots, x_n] = f[x_0, x_0, x_1, \ldots, x_n]$$

IX.

According to **Theorem 2.47**, the Chebyshev polynomial has the smallest maximum norm on $[-1, 1]$ among polynomials of degree n. namely,

$$\max_{-1 \leq x \leq 1} |\frac{T_n(x)}{2^{n-1}}| \leq \max_{-1 \leq x \leq 1} |p(x)|$$

Now, we want to find the similar polynomial $a_0 x^n + a_1 x^{n-1} + \cdots + a_n$ on $[a, b]$. So we need to transform $x \in [a, b]$ to $t \in [-1, 1]$:

$$t = \frac{b-a}{2} x + \frac{a+b}{2}$$

when $x = -1$, we have $t = a$; and when $x = 1$, we have $t = b$.

$$\therefore x = \frac{2t - a - b}{b - a}$$

Therefore, we have the wanted polynomial on $x \in [a, b]$:

$$a_0 x^n + a_1 x^{n-1} + \cdots + a_n = a_0 \frac{T_n(\frac{2x - a - b}{b - a})}{2^{n-1}}$$

Just sort out the coefficients of $x^n, x^{n-1}, \ldots, x^0$ and can simply get $a_0, a_1, \ldots, a_n$.

X.

We need to prove that

$$\max_{-1 \leq x \leq 1} |\frac{T_n(x)}{T_n(a)}| \leq \max_{-1 \leq x \leq 1} |p(x)|$$

Proof by contradiction: Suppose that

$$\exists p \in \mathbb{P}_n^a \quad s.t. \quad |\frac{T_n(x)}{T_n(a)}| > \max_{-1 \leq x \leq 1} |p(x)|$$

and

$$\because |T_n(x)| \leq 1, \quad x \in [-1, 1],$$

therefore, we only need

$$\exists p \in \mathbb{P}_n^a, \quad s.t. \quad |\frac{1}{T_n(a)}| > \max_{-1 \leq x \leq 1} |p(x)|. \tag{1}$$

Construct a new polynomial $Q(x) = T_n(x) - T_n(a)p(x)$, whose degree is at most n.

Since $T_n(x)$ has $n+1$ extremal points in $[-1, 1]$, denoted as $x_0^{'}, x_1^{'}, \ldots, x_n^{'}$, and at these points $|T_n(x_i)| = 1$.

So we have

$$Q(x_k^{'}) = T_n(x_k^{'}) - T_n(a)p(x_k^{'}) = (-1)^k - T_n(a)p(x_k^{'}), \quad k = 0, 1, \ldots, n.$$

When the k is even, due to **inequality 1** we have $Q(x_k^{'}) = 1 - T_n(a)p(x_k^{'}) > 0$; when the k is odd, we have $Q(x_k^{'}) = -1 - T_n(a)p(x_k^{'}) < 0$.

Since there are $n+1$ extrema, $Q(x)$ has at least n zeros in $[-1, 1]$. While $p \in \mathbb{P}_n^a$, so $Q(a) = T_n(a) - T_n(a)p(a) = 1 - 1 = 0$. In total, $Q(x)$ has at least $n+1$ zeros in $[-1, 1]$, and the degree of $Q(x)$ is at most n. Therefore, $Q(x) \equiv 0$, namely $\dfrac{T_n(x)}{T_n(a)} \equiv p(x)$, which is contradictory to **inequality 1**.

XI.

The Bernstein base polynomial is defined as

$$b_k^n(t) = \binom{n}{k} t^k (1-t)^{n-k}, \quad k = 0, 1, \dots, n.$$

The Conbinatorial number $\binom{n}{k}$ has the property that $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. Based on this property, we have:

According to definition, $b_{n-1,k}(t) = \binom{n-1}{k} t^k (1-t)^{n-1-k}$.

$b_{n,k}(t) = \binom{n}{k} t^k (1-t)^{n-k}$. $\therefore \frac{n-k}{n} b_{n,k}(t) = \binom{n-1}{k} t^k (1-t)^{n-k}$.

$b_{n,k+1}(t) = \binom{n}{k+1} t^{k+1} (1-t)^{n-1-k}$ $\therefore \frac{k+1}{n} b_{n,k+1}(t) = \binom{n-1}{k} t^{k+1} (1-t)^{n-1-k}$.

$\therefore RHS = \frac{n-k}{n} b_{n,k}(t) + \frac{k+1}{n} b_{n,k+1}(t) = \binom{n-1}{k} t^k (1-t)^{n-k} + \binom{n-1}{k} t^{k+1} (1-t)^{n-1-k}$

$= \binom{n-1}{k} t^k (1-t)^{n-1-k} \left[(1-t) + t\right] = \binom{n-1}{k} t^k (1-t)^{n-1-k} = b_{n-1,k}(t) = LHS.$

Figure 8: Problem XI

XII.

The Bernstein base polynomial is defined as

$$b_k^n(t) = \binom{n}{k} t^k (1-t)^{n-k}, \quad k = 0, 1, \dots, n.$$

$$\therefore \int_0^1 b_k^n(t)\, dt = \binom{n}{k} \int_0^1 t^k (1-t)^{n-k}\, dt$$

According to the definition of Beta function, we have

$$B(p,q) = \int_0^1 t^{p-1} (1-t)^{q-1}\, dt$$

Let $p = k+1, \quad q = n-k+1$. And Beta function has the property that

$$B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

For $k \in \mathbb{Z}, \quad \Gamma(k) = (k-1)!$.

$$\therefore \int_0^1 b_k^n(t)\, dt = \binom{n}{k} \frac{\Gamma(k+1)\Gamma(n-k+1)}{\Gamma(n+2)} = \binom{n}{k} \frac{k!(n-k)!}{(n+1)!} = \frac{1}{n+1}$$